

# Brain-Computer Interface (BCI) Motor Imagery Classification

Full End-to-End Pipeline Report  
EDA • Processing • Modeling • Deployment

|                         |                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Project Goal</b>     | Classify whether a person is imagining LEFT or RIGHT hand movement using EEG signals only (no actual movement). |
| <b>Data Size</b>        | ~2000 Excel files (trials). Each file = one trial with multi-channel EEG + label (LEFT / RIGHT).                |
| <b>Typical Channels</b> | FZ, C3, CZ, C4 (sensorimotor cortex + midline). More channels possible.                                         |
| <b>Core Pipeline</b>    | Load → Process → EDA → Feature Extraction / Deep Learning → Classify → Deploy                                   |
| <b>Target Users</b>     | Research team + future deployment for assistive technology demos                                                |

This document gives the complete roadmap from raw Excel EEG files to a working web demo. It is written for a small team so that each member knows what to do and why.

## Data Link:

<https://drive.google.com/drive/folders/128V1Hbfmw7S5tHZ4QLyvITZdKWk-Lj6G?usp=sharing>

## 1. Dataset Description

You have approximately **2000 Excel files**. Each file represents one experimental trial:

- Multiple columns = EEG channels (e.g. FZ, C3, CZ, C4, ...)
- Many rows = time samples (sampling rate typically 128–256 Hz)
- One label per file: **LEFT** or **RIGHT** (the class we want to predict)
- Duration per trial: usually 3–8 seconds of continuous recording

Important: Keep a clean mapping (filename → label). Store labels in a separate CSV or in the filename itself for easy loading.

## 2. Data Processing Pipeline

Raw EEG is very noisy. The following ordered steps turn the Excel files into clean signals ready for analysis and modeling.

| Step | Action              | Why it matters                                                                               |
|------|---------------------|----------------------------------------------------------------------------------------------|
| 1    | Load & Organize     | Read all ~2000 Excel files, attach LEFT/RIGHT labels, verify sampling rate and channel names |
| 2    | Band-pass Filter    | Keep 0.5–40 Hz (or tighter 8–30 Hz). Removes slow drifts and high-frequency noise            |
| 3    | Notch Filter        | Remove 50 Hz or 60 Hz power-line interference                                                |
| 4    | Artifact Rejection  | Drop trials/channels with extreme amplitude (eye blinks, muscle, electrode pops)             |
| 5    | Re-referencing      | Common Average Reference (CAR) or Laplacian to reduce shared noise                           |
| 6    | Epoching            | Cut the relevant time window after the cue (example: 0.5–3.5 s)                              |
| 7    | Baseline Correction | Subtract pre-stimulus mean so signals are centered                                           |
| 8    | Normalization       | Optional channel-wise or global scaling (helpful for deep learning)                          |

Table 1 — Recommended EEG Preprocessing Steps

### Key physiological target

Motor imagery mainly changes the **mu (8–12 Hz)** and **beta (13–30 Hz)** rhythms over the sensorimotor cortex (especially channels C3 and C4). Keeping these bands clean is the most important goal of processing.

### 3. Exploratory Data Analysis (EDA)

EDA answers the questions that decide whether the data is usable and which models are likely to work.

#### 3.1 Must-do checks

- Class balance: How many LEFT vs RIGHT trials? (aim for roughly equal)
- Sampling rate & trial length consistency across the 2000 files
- Missing values, NaNs, flat channels, extreme outliers
- Percentage of trials rejected by artifact rules (document it!)

#### 3.2 Visualizations that give insight

- Time-domain plots of a few LEFT and RIGHT trials on C3 and C4 (raw + filtered)
- Average Power Spectral Density (PSD) for LEFT vs RIGHT on C3 and C4
- Look for Event-Related Desynchronization (ERD): power decrease in mu/beta over the contralateral motor cortex
- Simple band-power comparison per channel (already hints which electrodes are informative)

#### 3.3 Why EDA matters for the team

If the PSD already shows a clear LEFT–RIGHT difference in the mu/beta range on C3/C4, then classical methods (CSP + SVM) and deep models both have a good chance. If no difference appears, go back to processing or data quality before modeling.

### 4. Modeling Strategy

Start simple and strong, then add more complex models for comparison. Recommended order:

| Priority     | Model                   | Input                                             | Strength                                | Notes for team                                     |
|--------------|-------------------------|---------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| 1 (Baseline) | CSP + SVM               | Log-variance features after CSP                   | Strong classic method for Motor Imagery | Implement first. Easy to interpret.                |
| 2            | EEGNet                  | Epoch multi-channel EEG (raw or lightly filtered) | Designed specifically for EEG           | Good deep-learning starting point, few parameters. |
| 3            | CNN / CNN-LSTM          | Same as EEGNet or spectrograms                    | Can capture more complex patterns       | Use if EEGNet is not enough.                       |
| 4 (Optional) | Transformer / Attention | Sequence of EEG or features                       | Modern, powerful                        | Only if time and data allow.                       |

Table 2 — Recommended Modeling Roadmap

#### Input / Output reminder

- **Input:** Clean multi-channel EEG (or CSP features)
- **Output:** LEFT or RIGHT (and preferably probability scores)

#### Evaluation metrics the team should report

Accuracy, Balanced Accuracy, F1-score, Confusion Matrix, and (if possible) AUC. Always use subject-aware or trial-aware cross-validation so the model is not tested on the same person's data it was trained on when possible.

## 5. Complete End-to-End Pipeline

| Stage            | Main Activities                                | Deliverable                        |
|------------------|------------------------------------------------|------------------------------------|
| 1. Data          | Collect / organize ~2000 Excel trials + labels | Clean file list + label map        |
| 2. Processing    | Filter, clean artifacts, epoch, baseline       | Clean epoched arrays (.npy / .pkl) |
| 3. EDA           | Plots, PSD, class balance, quality report      | EDA notebook + short report        |
| 4. Features / DL | CSP features or prepare tensors for EEGNet     | Feature matrix or DataLoaders      |
| 5. Modeling      | Train CSP+SVM, EEGNet, compare                 | Trained models + metrics           |
| 6. Evaluation    | Cross-validation, confusion matrices           | Final performance numbers          |
| 7. Deployment    | Wrap best model in a web app                   | Working demo (Streamlit / Flask)   |

Table 3 — Full Project Pipeline (highlight: EDA and Deployment)

## 6. Deployment Suggestions & Hints

After you have a trained model that reaches acceptable accuracy, the next step is to make it usable by non-programmers (demo for supervisors, possible future patients, or team presentations).

### 6.1 Which framework to choose?

| Framework | Best for                                   | Pros                                                | Cons / When to avoid                                 |
|-----------|--------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| Streamlit | Fast demo & internal tools                 | Very fast to build, pure Python, great for ML demos | Less suitable for complex multi-user production apps |
| Flask     | Lightweight custom web API + simple UI     | Full control, easy to add REST endpoints            | You write more HTML/JS yourself                      |
| Django    | Larger applications, admin, auth, database | Batteries-included, strong for multi-user systems   | Heavier; overkill for a first demo                   |

Table 4 — Deployment Framework Comparison (Streamlit recommended for first demo)

### 6.2 Practical deployment hints for the team

- **Start with Streamlit** — one Python file can load the model, accept an uploaded Excel/CSV trial (or a pre-processed array), run inference, and show LEFT/RIGHT + confidence.
- Save the final model with joblib (sklearn) or torch.save / onnx (deep models).
- Keep preprocessing code identical between training and inference (same filters, same window).
- Add a simple “example trial” button so judges/supervisors can try without uploading files.
- Later, if you need a public API or multi-user system, wrap the same model behind a Flask or FastAPI endpoint and optionally build a small frontend.
- For real-time future work: the same model can be called on sliding windows of live EEG (more advanced).

## 7. Suggested Team Division & Tips

| Role / Focus      | Main Tasks                                           | Key Deliverables                     |
|-------------------|------------------------------------------------------|--------------------------------------|
| Data & Processing | Load 2000 files, filtering, artifact rules, epoching | Clean dataset + processing notebook  |
| EDA & Quality     | Plots, PSD, statistics, rejection report             | EDA report / notebook                |
| Classical ML      | CSP + SVM pipeline, cross-validation                 | Baseline accuracy + confusion matrix |
| Deep Learning     | EEGNet (and optional CNN/LSTM)                       | DL model + comparison table          |
| Deployment        | Streamlit (or Flask) demo + model packaging          | Working web demo                     |
| Documentation     | Keep this pipeline updated, README, presentation     | Final report + slides                |

Table 5 — Suggested Team Responsibilities

### Important team tips

- Agree on a common folder structure and naming convention for the 2000 files early.
- Share a single “config” file (sampling rate, filter bands, epoch window, random seed).
- Use the same train/validation/test split (or same CV folds) for every model so results are comparable.
- Document every rejected trial and the reason — this is part of scientific honesty.
- Version the final model and the exact preprocessing code that produced the features it expects.

## 8. Summary & Next Actions

You have ~2000 labeled EEG trials (LEFT vs RIGHT). The path to a working system is:

1. Clean and standardize the signals (processing).
2. Understand them with solid EDA (especially PSD on C3/C4).
3. Build a strong baseline with CSP + SVM.
4. Add EEGNet (and optionally deeper models) for comparison.
5. Package the best model into a simple Streamlit (or Flask) demo so anyone can try it.

Focus first on data quality and a reliable baseline. A clean CSP+SVM that reaches good accuracy is more valuable than a complex model trained on noisy data. Deployment can start as soon as you have one trustworthy model — it does not have to wait until every experiment is finished.

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End of Full Pipeline Report

BCI Motor Imagery Classification — EDA • Processing • Modeling • Deployment